

CVAnalyser

AI-Powered CV/Resume Management System

Project Documentation

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1. Project Overview

What is CVAnalyser?

CVAnalyser is an AI-powered recruitment assistance platform that revolutionizes CV/resume management and candidate search. The system intelligently processes, stores, and enables natural language querying of candidate information through an advanced multi-database architecture combining graph databases, vector search, and multiple AI models.

Key Capabilities

- Parse and extract structured information from CVs in multiple formats (PDF, DOCX, TXT, images)
- Store candidate data across specialized databases (Neo4j graph, MongoDB, Pinecone vectors)
- AI-powered chatbot interface for natural language candidate queries
- Semantic and structured search through resume databases
- Bulk CV processing with concurrent operations
- Role-based access control and user management

2. Who Uses CVAnalyser?

Recruitment Teams

HR professionals and recruiters who manage large volumes of candidate CVs. They benefit from automated parsing, intelligent search, and the ability to quickly filter candidates based on skills, experience, and qualifications.

Hiring Managers

Decision-makers who need rapid access to candidate information through intuitive natural language queries such as 'Find candidates with Python skills and 5+ years experience' or 'Show me software engineers from top tech companies'.

System Administrators

IT staff responsible for managing user accounts, system prompts, and overall platform configuration with admin-level access to maintain and customize AI behavior.

Common Use Cases

- Bulk CV upload with automatic data extraction and structuring
- Natural language queries for candidate discovery
- Semantic search through resume contents for contextual matching

- CV management operations (view, delete, update, download)
- User account and permission management
- AI prompt customization for recruitment scenarios

3. Key Features

- **Multi-Format Support:** Process CVs in PDF, DOCX, TXT, and image formats
- **Natural Language Queries:** Chat-based interface for conversational candidate search
- **Semantic Search:** Vector similarity search for contextual matching
- **Graph Queries:** Complex relationship queries across candidate data
- **Bulk Processing:** Concurrent CV upload and processing with progress tracking
- **Real-time Chat:** Interactive interface with markdown support and history
- **Role-Based Access:** Admin and user roles with different permissions
- **Intelligent Caching:** Multi-level caching for optimal performance
- **Responsive Design:** Mobile-friendly with dark mode support

Summary

CVAnalyser is a production-ready AI recruitment platform that combines graph databases, vector search, and multiple LLM providers to create an intelligent candidate management system. The multi-database architecture ensures optimal performance for different query types, while the AI-powered natural language interface makes candidate discovery intuitive and efficient for recruitment teams, hiring managers, and HR professionals.